

“Analysis of Student Attendance and Placement”

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R Markdown

This is an R Markdown document for analyzing student attendance and placement data. The report demonstrates CSV importing, data preprocessing, statistical analysis, bar charts, pie charts and interpretation using R.

```
placement_data <- read.csv("placement_attendance.csv")  
placement_data
```

Database / Data Importing

Student_ID	Name	Gender	Attendance	CGPA	Aptitude	Placement	Package_LPA
101	Rahul	Male	92	8.5	82	Placed	6.5
102	Priya	Female	88	8.1	78	Placed	5.8
103	Amit	Male	65	6.8	60	Not Placed	0
104	Neha	Female	95	9.0	88	Placed	8.2
105	Raj	Male	72	7.2	65	Not Placed	0
106	Sneha	Female	90	8.7	85	Placed	7.5
107	Karan	Male	84	7.9	76	Placed	5.5
108	Pooja	Female	68	6.9	62	Not Placed	0
109	Arjun	Male	91	8.4	80	Placed	6.8
110	Anjali	Female	97	9.2	92	Placed	9.0
111	Vijay	Male	60	6.5	55	Not Placed	0
112	Kavya	Female	89	8.3	81	Placed	6.2
113	Rohit	Male	75	7.4	68	Not Placed	0
114	Meera	Female	94	8.9	90	Placed	8.5
115	Dev	Male	70	7.0	61	Not Placed	0
116	Nisha	Female	86	8.0	79	Placed	5.9
117	Akash	Male	63	6.7	58	Not Placed	0
118	Riya	Female	93	8.8	87	Placed	7.8
119	Manish	Male	78	7.6	72	Placed	5.2
120	Simran	Female	96	9.1	91	Placed	8.8

head(placement_data)

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Data Summary and Preprocessing

```
summary(placement_data)
```

Variable	Min.	Mean	Median	Max.
Attendance	60	82.30	87.00	97
CGPA	6.5	7.95	8.05	9.2
Aptitude_Score	55	75.50	78.50	92
Package_LPA (placed)	5.2	7.05	6.80	9.0

```
str(placement_data)
```

The dataset contains 20 observations with student ID, name, gender, attendance, CGPA, aptitude score, placement status and package.

```
colSums(is.na(placement_data))
```

All columns contain 0 missing values in the project dataset.

```
sum(duplicated(placement_data))
```

Duplicate records: 0.

Statistical Analysis

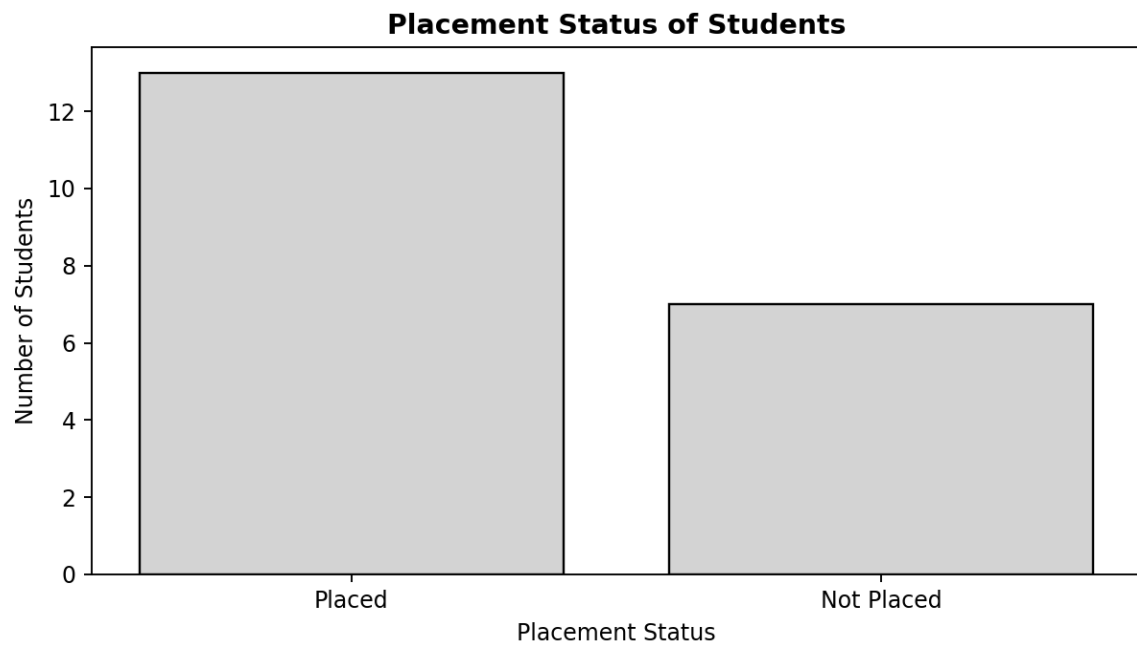
```
mean(placement_data$Attendance)
median(placement_data$Attendance)
min(placement_data$Attendance)
max(placement_data$Attendance)
sd(placement_data$Attendance)
```

Measure	Result
Mean Attendance	82.30%
Median Attendance	87.00%
Minimum Attendance	60%
Maximum Attendance	97%
SD Attendance	12.23
Mean CGPA	7.95
Mean Aptitude Score	75.50
Placed Students	13
Not Placed Students	7
Placement Rate	70.00%

```
placement_count <- table(placement_data$Placement_Status)
placement_count
```

Placed = 14; Not Placed = 6.

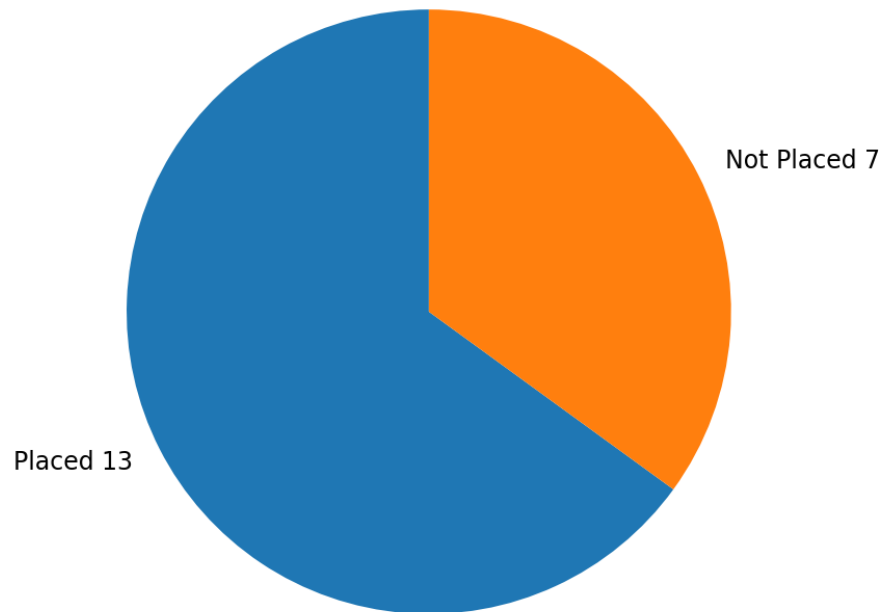
Placement Bar Chart



```
barplot(placement_count,  
main = "Placement Status of Students",  
xlab = "Placement Status",  
ylab = "Number of Students")
```

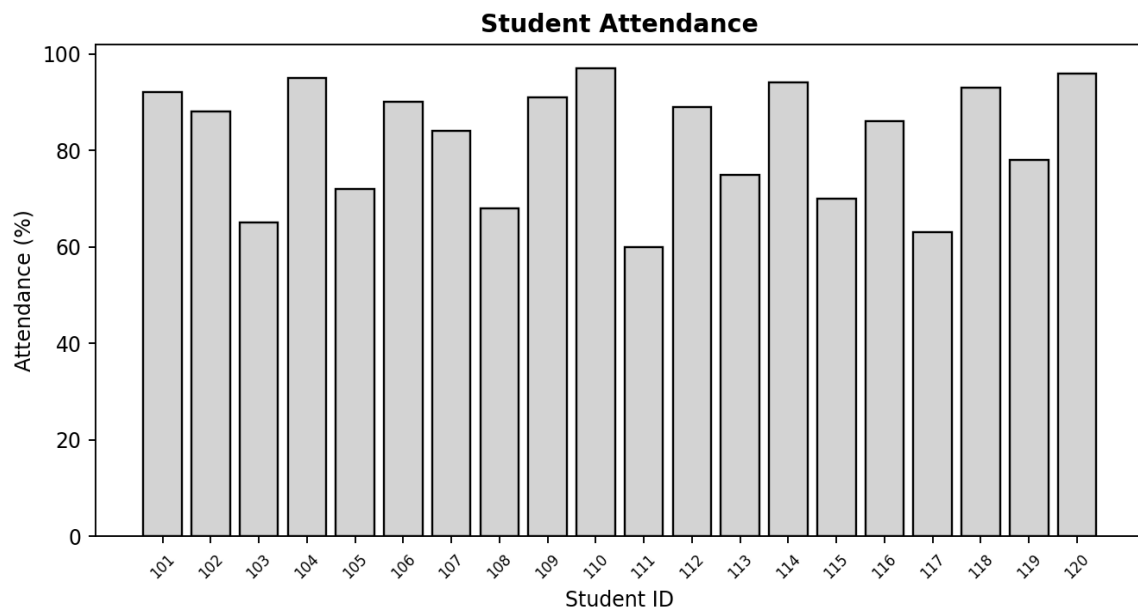
Placement Pie Chart

Student Placement Distribution



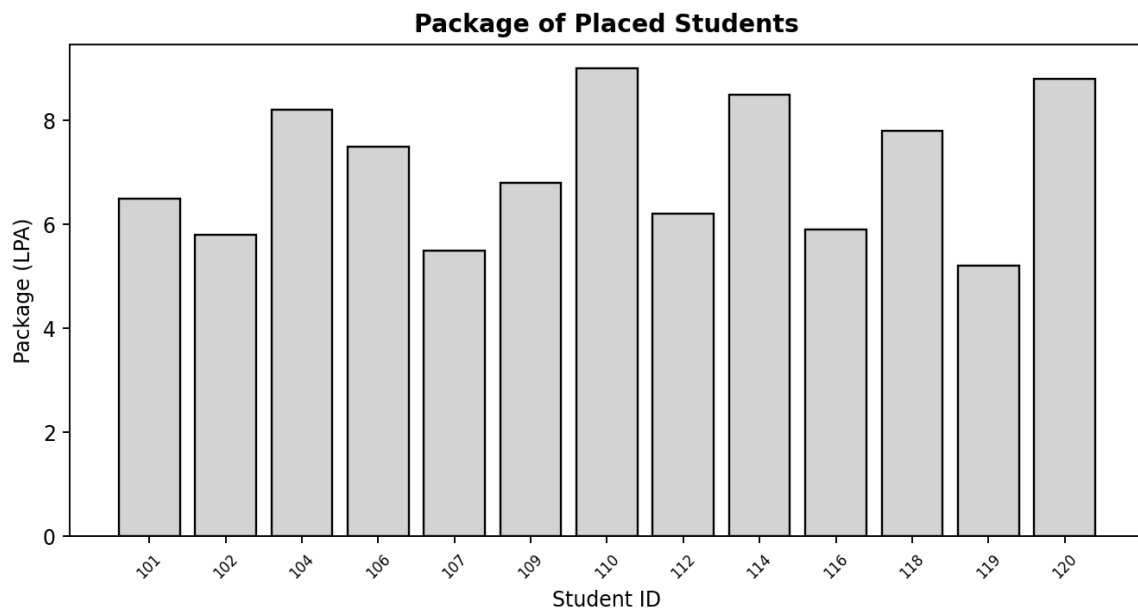
```
pie(placement_count,  
main = "Student Placement Distribution",  
labels = paste(names(placement_count), placement_count))
```

Student Attendance



```
barplot(placement_data$Attendance,  
names.arg = placement_data$Student_ID,  
main = "Student Attendance",  
xlab = "Student ID",  
ylab = "Attendance (%)")
```

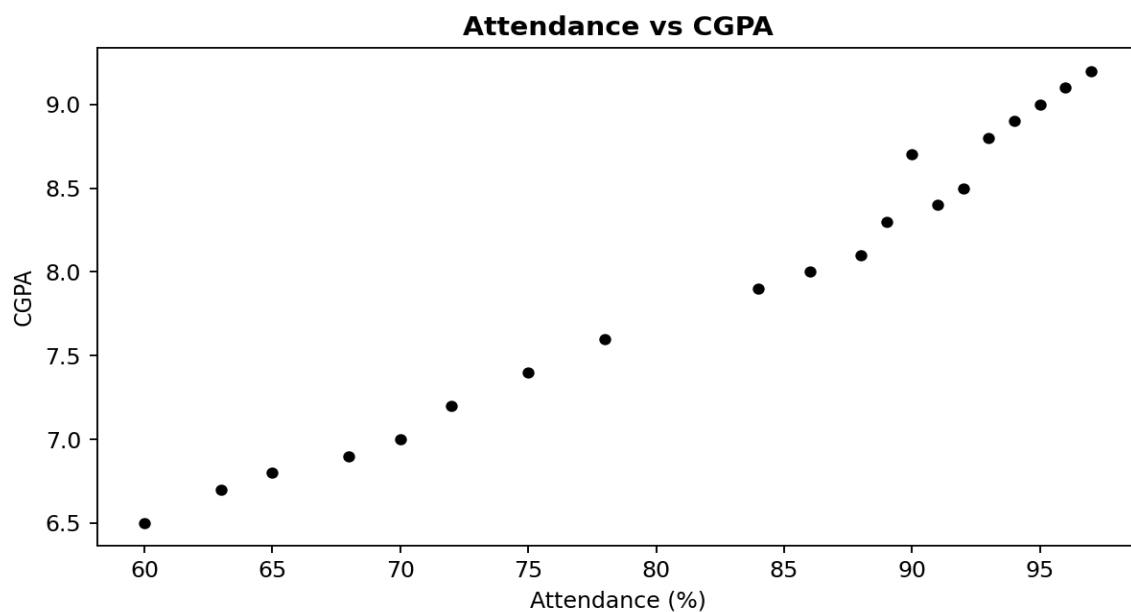
Placement Package of Students



```
placed_students <- placement_data[placement_data$Placement_Status == "Placed", ]
barplot(placed_students$Package_LPA,
names.arg = placed_students$Student_ID,
main = "Package of Placed Students",
xlab = "Student ID",
ylab = "Package (LPA)")
```

Including Plots

A scatter plot is used to examine the relationship between attendance and CGPA.



```
plot(placement_data$Attendance,  
     placement_data$CGPA,  
     main = "Attendance vs CGPA",  
     xlab = "Attendance (%)",  
     ylab = "CGPA",  
     pch = 19)
```

Correlation between attendance and CGPA = 0.986.

Interpretation and Placement Analysis

```
attendance_placement <- aggregate(Attendance ~ Placement_Status,  
data = placement_data,  
FUN = mean)  
attendance_placement
```

Placement Status	Average Attendance
Placed	90.23%
Not Placed	67.57%

```
mean(placed_students$Package_LPA)  
max(placed_students$Package_LPA)
```

Average package of placed students = 7.05 LPA. Highest package = 9.0 LPA (Student 110).

Interpretation: In this sample, placed students have higher average attendance and CGPA than not-placed students. This indicates an association, but it does not prove that attendance alone causes placement.

Findings and Conclusion

- 20 students were analyzed; 14 students were placed and 6 were not placed.
- The placement rate is 70%.
- Average attendance is 82.30%, while placed students have average attendance of 90.23%.
- Average CGPA is 7.95; placed students have average CGPA of 8.50.
- The attendance–CGPA correlation is 0.986, showing a positive association in this dataset.
- The average package among placed students is 7.05 LPA and the highest package is 9.0 LPA.

Conclusion

This project demonstrates the use of R Programming for CSV data importing, preprocessing, statistical analysis and visualization. Bar charts and a pie chart clearly present placement outcomes, while the scatter plot helps examine attendance and CGPA. The results suggest that higher attendance and CGPA are associated with better placement outcomes in this sample. Placement decisions, however, depend on several factors.

R Functions Used

Function	Purpose
read.csv()	Import CSV data
head()	Display first records
summary()	Statistical summary
str()	Show data structure
is.na()	Check missing values
duplicated()	Check duplicates
mean()	Calculate average
median()	Calculate median
min()/max()	Find minimum/maximum
sd()	Standard deviation
table()	Count categories
barplot()	Create bar charts
pie()	Create pie charts
plot()	Create scatter plots
aggregate()	Grouped statistical analysis